Computational Exploration of Microstructural Evolution in a Medium C-Mn Steel and Applications to Rod Mill

P. A. Manohar, Kyuhwan Lim, A. D. Rollett and Youngseog Lee*

Department of Materials Science and Engineering, Carnegie Mellon University,

5000 Forbes Avenue, Pittsburgh, PA – 15213, USA

*Plate and Rod research group, POSCO Technical Research Laboratories,

Pohang, P.O. Box 36, KOREA

Abstract

An ‘Expert System’ is proposed in this work to conduct computational exploration of the deformation and restoration behaviour of a medium C-Mn steel under high strain rate conditions, at elevated temperatures and complex strain paths that occur in rod rolling process. The expert system computes appropriate thermomechanical parameters necessary for describing rod rolling process in detail and then utilizes these parameters in mathematical models to determine microstructure evolution during a typical industrial-scale rod rolling process. Microstructure simulation in rod rolling is a challenging problem due to the fact that several softening processes may operate sequentially or concurrently during each deformation step. Different softening processes have very different impact on microstructure development and therefore it is important to investigate the particular combinations of processing conditions under which transition of operating softening processes occurs. In the present work, the transition from dynamic to metadynamic recrystallization is studied in detail based on the criteria of critical strain, austenite grain size and Zener-Hollomon parameter when the interpass (interdeformation) time is very short of the order of few milliseconds during the later stages of rod rolling. Computational results are subsequently validated by comparing the program output to in-plant measured microstructure data. The proposed expert system is designed as an off-line simulation tool to examine and assess the various options for thermomechanical process optimization.

Keywords: rod rolling process, microstructural evolution, mathematical modeling, thermomechanical processing, recrystallization kinetics.

1. Introduction

Contemporary objectives for steel processing are increasingly complex and often contrasting i.e. obtaining a desired product with optimum combination of properties such as strength, toughness, weldability and formability at lower cost. An important step to achieve these objectives is to develop mathematical models for the determination of deformation and recrystallization behaviour during thermomechanical processing (TMP) based on laboratory-scale hot torsion or compression tests. Computer simulations incorporating these models are then developed to determine optimum process sequences to obtain fine grained steels with desirable combination of micro constituents by controlling the process parameters such as strain, strain rate, temperature and cooling conditions in the strip and/or plate mill. However, application of such models to rod rolling requires extra care as this process is characterized by continuous multi-pass deformation (up to 31 passes) at high strain rate in the range of 0.4 – 3000s^{-1}, at elevated temperatures in the range 1173 – 1373 K, and very short interpass times of the order of 0.015 – 1.0*s*. These processing conditions make it virtually impossible to study experimentally the microstructural evolution both on laboratory and industrial scales. Hence, numerical and computational exploration remains to be the most important way to gain an insight in to the microstructure evolution during rod rolling. The knowledge of the in-process microstructural evolution is critical for both the optimization of the manufacturing schedule and adjustment of properties of the as rolled product [1,2)]. For example, a fine austenite grain size is desirable at the end of rolling to decrease its hardenability and to obtain a fine ferrite + pearlite structure *via* controlled rolling and cooling to eliminate or reduce the necessity of post-rolling annealing treatments such as annealing or spheroidization where required and to improve the mechanical

properties of the as-rolled products. Another example is reducing surface decarburization of high carbon spring steels and bearing steels by controlling the rolling process parameters and accelerated water-cooling before they enter the STELMOR air-cooling process

In these situations, determination of high temperature microstructural evolution through mathematical model becomes imperative. Previous efforts [3-5] to simulate microstructural evolution for rod rolling of medium C-Mn steels have concentrated mainly on calculating evolution of the mean austenite grain size. However, the relative contribution of dynamic and metadynamic recrystallization processes towards final microstructure has not been discussed in these reports. In addition, the calculation of effective true strains and strain rates in each deformation pass has not been detailed. In the present work, the focus is on several fundamental aspects of microstructure development including static, dynamic and metadynamic recrystallization processes and their kinetics, how to resolve the mechanism transitions when they operate concurrently and finally their specific contribution to microstructure evolution in continuous rod rolling process. It will be shown that the relevant concepts to determine the mechanism transitions as implemented in the proposed expert system allow the calculation of the relative amounts of dynamically and metadynamically recrystallized material during any deformation pass as a function of known values of process variables such as strain, strain rate, temperature, interpass time and in-process and post-process cooling conditions. In addition, pass-by-pass strain, strain rate and temperature calculation procedures are implemented as independent calculation sub-modules. Finally, the results from these sub-modules are combined in the expert system along with the metallurgical models to calculate microstructure evolution during the thermomechanical processing. Initial (i.e. as-reheated) microstructure, roll pass

schedule and cooling condition are the basic inputs to the system. Comparing the final predicted austenite grain size with that obtained in rod mill validates the model applicable for medium C-Mn steels. A further validation was possible by comparing the predicted ferrite grain size of the final product with that obtained *via* industrial processing.

2. Development of Expert System for Microstructural Evolution in Rod Rolling Process

The computation in the expert system begins by compiling the output from strain, strain rate and temperature calculation sub-modules implemented previously by one of the authors and co-workers [6-9] in to data arrays. The mathematical models incorporated in the expert system subsequently compute the microstructure evolution based on the computed data for process parameters during each pass. The program then determines the evolution of austenite grain size, fraction recrystallized and recrystallization kinetics for a given roll pass and cooling conditions. In-process grain growth is calculated subsequently. This process is iterated until the manufacturing process is finished. At this stage, the final microstructural parameters are computed using non-isothermal grain growth procedure. Finally, expected ferrite grain size is also determined. In the following sections, the pass-by-pass strain, strain rate and temperature sub-modules are described, subsequently details of the metallurgical model are given and finally, an outline of the whole expert system is presented.

2.1 Thermomechanical Modeling of Rod Rolling Process

a) Strain at a pass (pass-by-pass strain)

Rod rolling process involves a change in cross-sectional shape of the rolling stock through many sets of grooved rolls to obtain a final round shape of the rod from the original rectangular cross-section of the billet. The geometry of the roll grooves is non-rectangular (oval, round, diamond, square etc.) and therefore the workpiece undergoes complex strain and stress paths that cannot be simplified as either plane strain or plane stress. The average pass strains are typically calculated by multiplying the area strains, (i.e. the natural logarithm of the ratio of the fractional reductions in cross-sectional area through the rolling stands) by some constant factors [4)]. However, a mathematical rationale for the use of the constant factors has not been given. Another model [10)] for the calculation of pass strain and strain rate in rod rolling is based on the assumption of plane strain deformation. In this case, the three-dimensional deformation zone was subdivided into longitudinal strips of equal width in the roll gap direction and each strip was analyzed separately. No experimental verification was provided to support the assumption of plane strain deformation. In the present work, a model for calculating strain in rod and bar rolling proposed by one of the authors and coworkers [7-8)] is utilized. The model has been verified experimentally for rod and bar rolling. The model is based on the elementary theory of plasticity and the equivalent rectangle approximation method. Strain in a pass is assumed to be homogenously distributed across the stock. While the detailed calculation procedure is given in the cited works, the following main results are utilized in the present work.

The pass strain is defined as the maximum effective average strain at a given pass calculated according to the following relation:

$$\bar{\varepsilon} = \frac{2}{\sqrt{3}}\left[1+\left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2+\left(\frac{\varepsilon_1}{\varepsilon_2}\right)\right]^{1/2} \qquad \text{Eq. (1)}$$

Where $\bar{\varepsilon}$ is the pass strain, and ε_1, ε_2, and ε_3 [= -($\varepsilon_1 + \varepsilon_2$) in this case] are the plastic deformation components along the principal axes of deformation. These strain components are determined based on the dimensions of the rolling stock through following equations:

$$\varepsilon_1 = \ln\left(\frac{\overline{W}_i}{\overline{W}_p}\right) \qquad \text{Eq. (2)}$$

$$\varepsilon_2 = \ln\left(\frac{\overline{H}_i}{\overline{H}_p}\right) \qquad \text{Eq. (3)}$$

Where $\overline{W}_i$ and $\overline{W}_p$ are equivalent initial and final cross-sectional widths while $\overline{H}_i$ and $\overline{H}_p$ are equivalent initial and final cross-sectional heights of the work piece.

b) Strain rate at a pass (pass-by-pass strain rate)

Analogous to drawing and forging processes, the strain rate in rod rolling changes at various stages of deformation. The strain rate is the maximum at the entrance to the roll (or in its vicinity) and decreases along the roll bite, finally becoming zero at the outlet. For this reason, it

is necessary to introduce an *average* strain rate for a given pass. The *average* strain rate can be defined as the strain over a time interval taken for workpiece to pass through the roll gap. Hereafter the *average* strain rate is referred to as the strain rate. A critical question in the calculation of strain rate is how to define the effective roll radius at a given pass, which is a representative value of varying radius of roll groove. A procedure based on effective roll radius to calculate the pass-by-pass strain rate in rod rolling has been proposed in a previous publication [9]. According to this model, the mean strain rate is calculated through the following relation:

$$\bar{\dot{\varepsilon}} = \frac{\bar{\varepsilon}}{t_p} \qquad \text{Eq. (4)}$$

Where $\bar{\dot{\varepsilon}}$ is the average effective strain rate in a given pass and t_p is the time interval during which the stock resides in the roll-bite. The t_p is calculated according to the following equation:

$$t_p = \frac{60\bar{L}}{2\pi N R_{eff}} \qquad \text{Eq. (5)}$$

Where $\bar{L}$ is the effective projected contact length of the grooved roll with workpiece, N is the roll speed (rpm) and R_{eff} is the effective roll radius at a given pass.

c) Temperature evolution in rod and bar rolling

The temperature of the workpiece during rolling depends on various factors such as rolling speed, initial temperature of the billet, the amount and nature of the plastic deformation, the cross sectional shape of workpiece at each pass, cooling condition in the individual passes and distribution of cooling and equalization zone between stands (passes). To take care of the combined action of these parameters, a model for predicting temperature evolution of workpiece during rod and bar rolling can be formulated based on the following assumptions: i) uniform initial temperature of the billet; ii) no longitudinal temperature gradient (i.e., infinitely long rod); iii) uniform heat generation across the cross section of workpiece due to plastic deformation in the roll gap; and iv) circular cross sectional shape at each pass. Under these assumptions, temperature variation within the rod is governed by the one-dimensional axi-symmetric heat transfer equation [11)]. The solution method of the heat transfer equation using finite difference method with boundary conditions is as described in [12, 13)]. The values of conductivity and specific heat capacity of material have been taken to be a function of the temperature of the material and were obtained from the literature [14)]. To predict the temperature variation of workpiece under the action of water-cooling in the cooling zone and equalization zone between stands, the model proposed by Morales *et al*.[15, 16)] has been adopted. The model includes the influence of operation parameters such as rod size, rolling speed, rod temperature and water flow-rate on the temperature distribution within the rod under the action of water-cooling and simulated the temperature evolution of rod and bar for many cases. Both radiative and convective heat transfer models were incorporated in the thermal model to calculate stock cooling during rolling and water-cooling. The emissivity of the steel was assumed to be 0.8. During deformation in the roll

gap, most of the mechanical energy is transformed into heat. This deformation heating is determined for each element at each time step using a mean flow stress obtained by integrating the relevant constitutive equation. It should be mentioned here that frictional heat was ignored in comparison to the heat due to deformation in the current work.

2.2 Modeling of Microstructure Evolution

As steel is deformed at high temperatures, several types of softening processes occur in steel to reduce the internal energy of the deformed metal. When the deformation strain is high enough, the deformed grains may be replaced by new, strain free grains *via* a process termed recrystallization (RXN). When recrystallization occurs subsequent to deformation, it is termed static recrystallization (SRX). Under certain circumstances, the recrystallization may occur concurrently with deformation, a process called dynamic recrystallization (DRX). In some cases dynamic recrystallization may initiate through nucleation, but may not proceed to completion during deformation. In these cases, the recrystallization is completed after deformation by the growth of dynamically nucleated grains. This is known as metadynamic or postdynamic recrystallization (MDRX). The conditions under which SRX, DRX and MDRX occur are determined by the combination of processing parameters.

a) RXN Phenomena

The particular combinations of process parameters that lead to different softening phenomena can be understood from the schematic high temperature stress – strain (σ - ε)

diagrams for steels shown in **Figure 1**. The stress-strain diagrams of the type shown in **Fig. 1** are obtained in various testing schemes such as uniaxial tension, uniaxial compression, torsion, and plane strain compression. The strain and stress values obtained in each type of tests may be converted to equivalent strain and equivalent stress and plotted as shown in **Fig. 1**, *via* a suitable yield criterion (e.g. von Mises) so that the flow behaviour obtained in different kinds of tests could be compared on a common platform. It may noted from **Fig. 1** that σ increases as ε increases in the initial part of the curves through work hardening up to a critical strain, ε_c, beyond which DRX is initiated. The material work hardens as strain increases further, but at a reduced rate, owing to dynamic recovery and the onset of DRX until ε reaches another critical value, ε_p, peak strain with a corresponding stress value, σ_p, peak stress. At this stage stress falls as the strain is increased beyond ε_p because the rate of work hardening is offset by the rate of softening owing to the significant amount of DRX. With further straining, material deforms under a steady state regime denoted by σ_{ss} - ε_{ss}. DRX may not reach completion during a deformation pass and the growth of dynamically nucleated nuclei would continue during the interpass leading to MDRX condition. When strain is less than ε_c, static recrystallization may be initiated during the interpass time.

b) Transition conditions for different RXN Phenomena

Mathematical models that describe the kinetics of the various recrystallization processes are given in **Table 1**. The different softening processes (SRX, DRX, MDRX) result in different high temperature microstructures in steels and therefore it is essential to determine which process(es) are active during a given deformation pass. One way to deal with the problem of

transition conditions that separate SRX and MDRX in strip rolling of steel was proposed by Sun and Hawbolt [24] where authors suggested that two conditions viz. $\varepsilon > \varepsilon_c$ and $Z < Z_{lim}$ must be met simultaneously for MDRX to occur, where Z_{lim} is the limiting value of the Zener-Hollomon parameter given according to:

$$Z_{lim} = 5x10^{15} \exp(-0.0155d_0), \quad \text{Eq. (6)}$$

Where d_0 is the austenite grain size prior to recrystallization. Such a model is empirically based and suggests a rather straightforward relation between transition conditions for SRX/MDRX with initial grain size. Also, the transition conditions for DRX/MDRX remain to be defined with this approach while they are most relevant to the rod rolling process. Kinetics of concurrent and competing events of strain accumulation and relaxation must decide the transition conditions for SRX, DRX and MDRX and therefore more detailed microstructure descriptors are required to define the transition conditions between the different softening phenomena. In addition, the process conditions in strip rolling are quite different (strain rates slower, interpass times longer and temperature range wider in strip rolling) compared to those in rod rolling and therefore the previously suggested transition conditions will not be applicable to rod rolling. Therefore, an alternative approach is proposed in this work to resolve the DRX and MDRX transition as described in the following section.

c) Peak strain, critical strain and recrystallized volume fraction model

In the current work, the kinetics of dynamic recrystallization described in terms of peak and critical strains were utilized to resolve DRX / MDRX transition. The peak and critical strains were calculated based on the following relations [22]:

$$\varepsilon_p = 6.97 \times 10^{-4} \times D_o^{0.3} \times Z^{0.17} \quad \text{Eq. (7)}$$

$$\varepsilon_c = 0.81 \times \varepsilon_p \quad \text{Eq. (8)}$$

Eq. (7) includes the effect of initial austenite grain size (controls diffusion path length), and temperature-compensated strain rate (Zener-Hollomon parameter) on strain accumulation and relaxation, which in turn decide the critical strain for the onset of DRX. In the strain range from ε_c and ε_p, DRX will initiate, but perhaps will not lead to completion when the interpass times are short. In such cases, the amount of fraction recrystallized dynamically is computed according to a version of JMAK kinetics given according to the following equation [17]:

$$F_{XDRX} = 1 - \exp\left[B\left(\frac{\varepsilon - \varepsilon_c}{\varepsilon_p} \right)^k \right] \quad \text{Eq. (9)}$$

d) Grain growth model

The austenite grain growth under continuous cooling conditions is calculated by using the additivity rule where the cooling thermal cycle was divided into isothermal time steps [5,25] through the following relation for MDRX grains [4]:

$$D_f^7 - D_o^7 = 8.2x10^{25} x \sum_i \delta t_i x \exp(-400000/RT_i), \qquad \text{Eq. (10)}$$

Where δt_i is the length of time step i and T_i is the temperature of step i.

Grain growth during the time that the stock cools from finish rolling temperature up to the $\gamma \rightarrow \alpha$ transformation start temperature is of critical importance that determines the final ferrite grain size along with the cooling rate through the $\gamma \rightarrow \alpha$ transformation temperature range. The ferrite grain size ($\alpha_{g.s.}$) prediction model developed for hot strip rolling [26] is adopted here:

$$\alpha_{g.s.} = \{V_\alpha \text{ x } \exp(50.7 \text{ x } D_f^{0.024} - 51000/Ar_3)\}^{(1/3)} \qquad \text{Eq. (11)}$$

Where V_α is the volume fraction of ferrite (determined experimentally in this work), D_f is the austenite grain size just prior to the onset of $\gamma \rightarrow \alpha$ transformation (μm), and Ar_3 is $\gamma \rightarrow \alpha$ transformation start temperature (^oC) under prevalent cooling conditions (1.5 K/s in the present case) for a given composition.

3. Applications to Rod Mill

A continuous rod rolling process of POSCO No. 3 rod mill is shown in **Fig. 2**. It consists of thirty one passes in all, however the last two passes are sizing passes and are ignored for

rigorous analysis in the present study. It is evident from **Fig. 2** that several delay zones exist in the thermomechanical processing sequence for example, between pass #5 and #6; pass #11 and #12 (not shown in **Fig. 2**); pass #13 and #14 and pass #19 and #20 while two cooling zones are employed between pass #27 and #28 and final cooling subsequent to pass #31 where the temperature is reduced from finish rolling temperature ~ 1243 K down to ~ 1103 K using high speed water cooling (~1000 K/s). Further cooling of the rod occurs in STELMOR cooling line at a much slower cooling rate of ~1.5 K/s. The rolling speed increases from 0.12 m/s at the first pass up to 110 m/s in the final pass as the 160 x 160 mm square billet is reduced to ~ 6.0 mm diameter rod. Area reductions obtained in each pass varies in the range 17% – 27%, except for the final two passes where it is around 5%. The last two passes (#30 and #31) are the final sizing passes and are ignored for microstructure modelling in this work.

4. Results

The flow chart for the expert system that simulates thermomechanical processing for rod mill is given in **Figure 3**. Initial (i.e. as-reheated) microstructure and rolling schedule are the basic inputs to the system. The program then calculates the deformation conditions such critical and peak strains, and Zener – Hollomon parameter based on initial grain size, strain, strain rate and temperature. The logical conditions described in **Section 2.2c** along with relevant mathematical models listed in **Table 1** are used to determine which RXN mechanism operates for the calculated combination of process parameters and to compute the evolution of grain size, fraction recrystallized and recrystallization kinetics for a given roll pass schedule and finally,

expected ferrite grain size is determined using **Eq. (11)**. The expert system is implemented on a PC workstation in Windows environment using object oriented C++.

4.1 Output of the expert system

Detailed microstructural evolution computed by the expert system is given in **Table 2**. The data given in **Table 2** show that significant dynamic recrystallization followed by MDRX occurs in the roughing mill (pass #2 - #10), while remaining passes of the roughing mill (#11 - #13) show MDRX as the main operating recrystallization mechanism. Intermediate finishing mill passes (#14 - #19) are characterized by either static or metadynamic recrystallization process being more dominant. The MDRX process also dominates in almost all passes of the finishing mill (#20 - #27). The last two passes of the reduction and sizing mill (#28 and #29) are interesting where stain rate is very high (> 2000 s^{-1}), interpass time very short (~ 9 – 12 *ms*), and relatively low temperature (~1243 K). In this case, critical strain is too high to initiate DRX in pass #28 so that SRX initiates in the interpass time. However, the time available for recrystallization is so short that only about 36% of the volume is recrystallized. This leads to a large amount of retained strain in the material so that the effective strain in pass #29 exceeds the critical strain despite a high value of ε_c leading to DRX followed by MDRX in this pass. The austenite grain size gets refined from an initial value of ~300 μm down to ~3.3 μm after finish rolling. The austenite grains coarsen to a size of ~15 μm during cooling from finish rolling temperature to the cooling stop temperature (CST), and subsequently during slower cooling from CST to Ar_3. This not surprising because the material studied is a medium C-Mn steel where the grain growth is not inhibited by second phase particles or through any significant solute drag

effect. Also, the grain size is very fine at this stage, which means that the grain growth rate is expected to be high according to the curvature driven grain growth theory [27]. This means that the advantage gained in refining the austenite grain size through DRX and MDRX during rolling is lost to an extent during the cooling segments. Evolution of mean austenite grain size and the corresponding strain rate as a function of rolling pass is shown in **Fig. 4**. Significant increases in the mean austenite grain size due to in-process grain growth during the TMP delays mentioned in **section 3** are also evident in **Fig. 4**.

4.2 Validation of expert system

The specimens obtained from POSCO No. 3 rod mill were sectioned, mounted, polished and etched in 4% Nital in the usual way. The micrographs in **Fig. 5a** were taken with a Philips SEM at 400X and 3000X to obtain the appropriate resolution for prior austenite grain size and ferrite grain size respectively. The images taken for measuring austenite grain size were traced along ferrite grains precipitated along prior austenite grain boundary to delineate prior austenite grains. It should be mentioned here that some experience and judgment is necessary in tracing the prior austenite boundaries. Better accuracy in measurement may be obtained from a fully martensitic microstructure by the use of well-known metallographic etching techniques to reveal more clearly the prior austenite grain boundaries. However, it is not possible to quench the rod in industrial processing to a fully martensitic microstructure prior to coiling. Despite this limitation, the overall statistics of the austenite grain size distribution as measured by this method is shown in **Figure 5c**. It is evident from **Fig. 5c** that the measured and normalized grain size distribution compares well with the lognormal distribution that is expected under post-processing normal

grain growth regime. The distribution of the measured austenite grain size is analyzed by non-linear curve fitting method using the following expression for the lognormal distribution:

$$f(x) = y_0 + \frac{A}{\sqrt{2\pi}wx}\exp\left[\frac{-\ln\left(\frac{x}{x_c}\right)^2}{2w^2}\right] \qquad \text{Eq. (12)}$$

Where, $f(x)$ is the probability density of the continuous, random, and independent variable x (i.e. area-equivalent austenite grain diameter in this case), y_0 and A are fitting coefficients, w is skewness and x_c is kurtosis of lognormal distribution. Skewness represents a lack of symmetry in a probability distribution while kurtosis is a measure of how "fat" a probability distribution's tails are. The R^2 value of the lognormal distribution curve fit shown in **Fig. 5c** is fairly close to 1, which means that the grain size measurement procedure described above is reasonable.

The average austenite grain size was subsequently determined using the Scion-Image image analysis program to measure area of each grain on the traced image and area equivalent diameter was calculated according to the procedure outlined in the relevant ASTM standard [28)]. Austenite microstructures are presented in **Figure 5a** and **5b** while ferrite microstructures are presented in **Figure 6a** and **6b**. The predicted austenite grain size just before the onset of $\gamma \rightarrow \alpha$ transformation is 15.2 μm while the measured prior austenite grain size in industrially processed steel is 14.4 μm. The transformed ferrite grain size is measured to be 4.6 μm while the predicted ferrite grain size is 4.9 μm. It is clear from this data that the predicted and experimentally

measured mean austenite and ferrite grain size compare quite well, which validates the expert system that simulates the microstructural evolution in rod rolling process.

5. Discussion

The close agreement between the predicted austenite grain size at the end of rolling and that obtained from industrially processed material is interesting. The mathematical models used in this work to compute the microstructural evolution had been determined in laboratory torsion testing under strain rate conditions of up to 100 s^{-1} while industrial rod rolling involves strain rates as high as 3000 s^{-1}. Constitutive relations in high strain rate regime could be different compared to those determined in laboratory under the low strain rate regime. This should have introduced certain inaccuracies in the computed microstructure. One possible reason why this has not happened can be explained by studying the parameters for the industrial process sequence simulated here. It is clear from the data presented in **Table 2** that the first seventeen passes have a strain rate ≤ 100 s^{-1}, which compares well with the strain rates used in laboratory studies. The austenite grain size reaches ~ 8 μm at this stage. Further processing of the material does not seem to refine the austenite grain size to any significant extent in this material. It should be noted here that the steel composition being studied is plain carbon steel that is characterized by rapid recrystallization and grain growth kinetics in the absence of any precipitate pinning or a strong solute drag caused by solutes such as Mo and Nb. As a result, the refinement of austenite grain size achieved during the last stages of processing is lost due to rapid grain growth via in-process and non-isothermal grain growth during cooling. For more complex compositions such

as low alloy and microalloyed steels, high strain rate constitutive relations would be required to be incorporated in the expert system.

Based on the foregoing analysis, the major operating mechanism as a function of austenite grain size and Zener-Hollomon parameter may be determined as shown in **Figure 7**. The three-dimensional plot shown in **Fig. 7** provides further insight in to particular combinations of process and microstructural variables for SRX, DRX and MDRX to occur. The points for MDRX in **Fig. 7** are clustered, which indicates that MDRX is the dominant softening mechanism when the austenite grain size is relatively small (3 – 17 μm) and Zener-Hollomon parameter is in the range 5 – 70 x 10^{14}. The DRX is dominant where the austenite grain size is in the range 20 – 40 μm and the Zener-Hollomon parameter is comparatively low in the range 0.02 – 0.5 x 10^{14}. The SRX dominates when pass strain does not exceed the critical strain for a given pass, as expected. It should be mentioned here that the data for the first pass is not included in **Fig. 7** as the initial austenite grain size in this case is very coarse as-reheated grain size ~ 300 μm and would not lead to complete DRX because of the long diffusion distances involved.

It is also noteworthy that the expert system developed in the current work predicts metadynamic recrystallization as the predominant grain refinement softening mechanism which agrees well with previous studies [20,29] based on laboratory simulation and industrial process data. Therefore the current model may now be used to analyze the capabilities of rod rolling process when the critical process variables are changed in the model. The data shown in **Figure 8a** demonstrates that a 50% reduction is austenite grain size obtained by post deformation accelerated cooling would yield a 20% refinement in ferrite grain size. It is well known that

further ferrite grain refinement is possible by increasing the cooling rate through the $\gamma \rightarrow \alpha$ transformation range. On the other hand, reducing the finish rolling temperature for existing cooling rates does not appear to have significant influence on austenite grain size prior to the $\gamma \rightarrow \alpha$ transformation (**Figure 8b**). The main reason for this behaviour is that the rapid grain growth kinetics in C-Mn steel during cooling segments offsets the advantage gained when the finish rolling temperature is reduced. Microalloy addition (e.g. Ti, Nb) to the steel composition would retard the post deformation grain growth significantly and thus it would be possible to retain most of the austenite grain refinement advantage accrued due to DRX and MDRX processes.

6. Conclusions

1. An expert system is proposed in this work to compute the microstructural evolution in rod rolling that involves high strain rate deformation and complex strain paths. The predicted austenite microstructure at the end of rolling and the ferrite grain size subsequent to transformation correlate well with the data obtained from industrially processed material thus validating the simulation procedure.

2. Transition between dynamic recrystallization (DRX) and metadynamic recrystallization (MDRX) has been resolved based on austenite grain size and Zener-Hollomon parameter. It is found that MDRX dominates when austenite grain size is comparatively fine (3 – 17 μm) and Zener-Hollomon parameter is relatively high ($5 – 70 \times 10^{14}$) compared to conditions for DRX ($D_o = 20 – 40$ μm, $Z = 0.02 – 0.5 \times 10^{14}$).

3. Two main techniques for austenite grain refinement in rod rolling appear to be the effective utilization of DRX and MDRX processes in the final passes of rod rolling and increasing cooling rates subsequent to rolling and through the $\gamma \rightarrow \alpha$ transformation range. Both of these techniques will result in significant ferrite grain refinement. However, the reduction in finish rolling temperature *per se* appears to have limited scope in this regard due to rapid grain growth kinetics during final cooling segments.

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